

Abhijit Shidlagatta

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PERSONAL PROFILE

A highly technical Computer Science graduate (2:1) specialising in C++ engine programming, real-time procedural generation, and custom GPU shader development. Proven ability to tackle complex performance challenges, including multithreading and memory optimisation, seeking programming or technical art roles to contribute to cutting-edge gameplay and rendering systems.

EDUCATION

City St George's, University of London

BSc (Hons) Computer Science

Sep 2023 – Jun 2026

Upper Second-Class Honours (2:1)

• **Key Modules:** Advanced Games Development, Object-Oriented Programming, Database Design, Data Structures.

Tiffin School

Completed 2023

A-Levels: Mathematics (A*), Physics (B), Economics (B) | **GCSEs:** 12 subjects including Maths (8) & English

TECHNICAL PROJECTS

Real-Time Procedural Ocean Renderer | C++, Unreal Engine 5, HLSL

2026

- Implemented Tessendorf's FFT ocean simulation from scratch as HLSL compute shaders, replacing Unreal's built-in Water plugin with a fully custom GPU spectral wave pipeline.
- Engineered a multi-cascade Cooley-Tukey radix-2 FFT (spectrum initialisation, dispersion evolution, butterfly passes, inversion) orchestrated through Unreal's Render Dependency Graph (RDG) for real-time GPU dispatch.
- Derived a physically-based water colour model from published absorption/scattering spectroscopy data (Pope & Fry, Jerlov), exposing runtime-switchable presets via Unreal's Single Layer Water shading model.

Dissertation: Procedural Destructible Voxel Environment (Grade: 86.2%) | C++, Unreal Engine 5 2025-2026

- Engineered a real-time procedural dungeon generator in C++ and UE5, balancing dynamic voxel destructibility with structured room layouts.
- Implemented Axis-Aligned Bounding Box (AABB) spatial querying, resulting in constant-time collision validation that prevents topological overlaps.
- Optimized heavy $O(n^3)$ voxel carving operations by offloading calculations to background worker threads via Unreal's AsyncTask API, maintaining a stable 60 FPS.

Action RPG & OpenGL Graphics Engine | C++, C#, OpenGL, Unity

2025-2026

- Built an OpenGL spline-track racing game from scratch, integrating custom bloom post-processing and core graphics API rendering techniques.
- Developed a combat-focused "Souls-like" Action RPG for a strict university deadline, architecting finite state-machine AI and frame-perfect hitbox logic.

WebGL ASCII Render Pipeline | GLSL, WebGL, Three.js

2026

- Engineered a custom, GPU-accelerated rendering pipeline using highly complex GLSL fragment shaders.
- Implemented advanced image processing mathematics, including a real-time 3x3 Sobel operator for directional edge detection and Bayer matrix ordered dithering.

Database-Linked Calendar System | Java, SQL

2024-2025

- Took ownership and successfully integrated a Java back-end interface with an SQL database.
- Managed strict version control and seamless code integration in an Agile student team to meet business requirements.

Indie Game Jams (GMTK) | Unity, Godot, C#, C++

2026

- Conceptualised, designed, and coded fully functional indie games within strict 48-hour time limits, focusing on rapid debugging and clean art-to-engine pipelines.

TECHNICAL SKILLS

Programming: C++, C#, Java, Python, JavaScript/TypeScript, GLSL/HLSL

Tools & Technologies: Git/GitHub, Unreal Engine, Unity, WebGL, OpenGL, Three.js, Blender

Core Competencies: Object-Oriented Programming (OOP), Graphics Programming, Multithreading, Memory Optimisation, Procedural Generation, Shader Development, Game Engine Architecture, Technical Art Pipelines, Digital and Traditional Illustration, Concept Art, 3D Modelling and Rendering, Level Design